Kayondo Ernest & Group 7

Complete Code Explanation & Presentation Guide • Makerere University

> **How to use this guide:** Keep this PDF open on your phone during your presentation or defense. It gives you plain, direct answers to explain every single file, tag, CSS rule, and JavaScript function to your lecturer with confidence.

1. Overview of the 6 Unique Website Archetypes

Our website represents six Makerere University students in Group 7. Each student designed their page using a completely different real-world website archetype, proving separate and individual design work:

- **ernest.html • Kayondo Ernest:** *Instagram Profile & Feed Layout* (Story Highlights for hobbies, posts feed, follower stats).
- **daniella.html • Kagoda Daniella Evelyn:** *Notion Digital Workspace* (Page properties, breadcrumb, Kanban board columns).
- **ibrahim.html • Lubega Ibrahim:** *LinkedIn Professional Profile* (Cover photo, headline, experience timeline for golf & football).
- **ntwari.html • Ntwari Joseph David:** *ESPN / Sports Match Center* (Large squad jersey #10, performance radar, match report).
- **index.html • Nantume Bakhita Grace:** *Modern Software Developer Portfolio* (Asymmetrical sidebar + content stream).
- **matthew.html • Musiinguzi Matthew Jekyns:** *Tactical Chess & Basketball Arena* (High-contrast dark court layout).

Key Shared Principles:

- **No Emojis:** All icons are standard vector icons loaded via Font Awesome CDN.
- **No Artificial Highlight Strips:** All cards have clean, uniform, balanced borders.
- **Individual Stylesheets:** Each student has their own CSS file (e.g. `css/ernest.css`), proving separate work.

2. Line-by-Line Breakdown: Kayondo Ernest's Webpage (ernest.html)

Section A: Document Header & Font Awesome Link

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Kayondo Ernest (@ernest.devops) • Makerere University</title>
  <link rel="stylesheet" href="https://cdnjs.cloudflare.com/ajax/libs/font-awesome/6.5
  <link rel="stylesheet" href="css/ernest.css">
</head>
```

How to explain this to the lecturer:

- `<!DOCTYPE html>`: Tells the browser to render the page using the latest HTML5 standard.
- `<meta charset="UTF-8">`: Sets standard character encoding for text.
- `<meta name="viewport" ...>`: Ensures the webpage adjusts nicely on mobile phones and tablets.
- `<link rel="stylesheet" ... font-awesome ...>`: Connects the external Font Awesome library so we can display clean icons (like the swimming, running, and gym icons) instead of emojis.
- `<link rel="stylesheet" href="css/ernest.css">`: Links Ernest's personal stylesheet with the Instagram UI theme.

Section B: Navigation Bar

```
<header class="ig-top-header">
  <div class="ig-top-container">
    <a href="index.html" class="ig-brand">Instagram <span>Group 7</span></a>
    <nav class="ig-nav-links">
      <a href="index.html">Grace</a>
      <a href="daniella.html">Daniella</a>
      <a href="ernest.html" class="active">Ernest</a>
      <a href="ntwari.html">Joseph</a>
      <a href="ibrahim.html">Ibrahim</a>
      <a href="matthew.html">Matthew</a>
    </nav>
  </div>
</header>
```

How to explain this to the lecturer:

- Styled like Instagram's sticky top header with logo on the left and navigation on the right.
- The link for Ernest has `class="active"`, which highlights his tab in Instagram blue (`#0095f6`).

Section C: Instagram Profile Header & Story Highlights

```
<section class="ig-profile-card">
  <div class="ig-profile-top">
    <div class="ig-avatar-ring">
      <div class="ig-avatar-inner"><i class="fa-solid fa-code"></i></div>
    </div>
    <div class="ig-meta-side">
      <div class="ig-username-row">
        <span class="ig-username">ernest.devops</span>
        <i class="fa-solid fa-circle-check ig-verified"></i>
        <button id="like-btn" class="ig-btn-primary">
          <i class="fa-solid fa-thumbs-up"></i> Follow & Like (<span id="like-count">0
        </button>
      </div>
      <div class="ig-stats-row">
        <div><span>3</span> posts</div>
        <div><span>256</span> followers</div>
        <div><span>128</span> following</div>
      </div>
      <div class="ig-bio">
        <h1>Kayondo Ernest</h1>
        <p class="tagline">Aspiring DevOps & Cloud Infrastructure Engineer</p>
        <p>Undergraduate Student at Makerere University, Kampala</p>
      </div>
    </div>
  </div>
  <!-- Story Highlights for Hobbies -->
  <div class="ig-stories-bar">
    <div class="ig-story-item">Swimming</div>
    <div class="ig-story-item">Running</div>
    <div class="ig-story-item">Gym</div>
    <div class="ig-story-item">Gaming</div>
  </div>
</section>
```

How to explain this to the lecturer:

- Uses the signature Instagram colorful gradient border ring around the avatar.
- Displays follower stats, verified checkmark, and student bio.

- Ernest's 4 genuine hobbies (Swimming, Running, Gym, Gaming) are displayed as circular Instagram Story Highlights.

Section D: Instagram Posts Feed

```
<div class="ig-posts-column">
  <!-- Post 1: Fitness & Morning Runs -->
  <article class="ig-post-card">
    <div class="ig-post-header">...</div>
    <div class="ig-post-body-visual">Morning Campus Jogs & Recovery Swimming</div>
    <div class="ig-post-actions"><i class="fa-solid fa-heart"></i> ...</div>
    <div class="ig-post-caption">
      <strong>ernest.devops</strong> Morning run around Makerere campus...
    </div>
  </article>
  <!-- Post 2: DevOps Engineering Journey -->
  <!-- Post 3: Evening Gaming Sessions -->
</div>
```

How to explain this to the lecturer:

- Rather than a plain text blog, Ernest's thoughts are formatted as real Instagram posts with user badge, action icons (Heart, Comment, Share), and captions.

Section E: Instagram Comments Box

```
<section class="ig-comments-card">
  <h2><i class="fa-solid fa-comments"></i> Leave a Comment for Ernest</h2>
  <form id="comment-form" class="ig-comment-form">
    <input type="text" id="comment-author" placeholder="Add your username or name..."
    <textarea id="comment-message" placeholder="Add a comment..." required></textarea>
    <button type="submit">Post</button>
  </form>
  <div id="comments-list" class="comments-list"></div>
</section>
```

How to explain this to the lecturer:

- The button has `id="like-btn"` and the counter has `id="like-count"`.
- When a comment is submitted, JavaScript intercepts the submit and inserts the comment into `#comments-list` without reloading the page.

3. How the JavaScript Works (js/main.js)

The JavaScript is lightweight (under 60 lines) and uses plain vanilla JavaScript.

1. DOMContentLoaded Listener

```
document.addEventListener('DOMContentLoaded', () => {
  const student = document.body.getAttribute('data-student') || 'student';
  ...
});
```

- Waits until all HTML elements are loaded before running the script so it never crashes.
- Reads `data-student="ernest"` so Ernest's likes and comments are stored in his own separate storage key.

2. The Like Button Logic

```
const storageKey = 'group7_likes_' + student;
let count = parseInt(localStorage.getItem(storageKey), 10) || 0;
likeCountSpan.textContent = count;

likeBtn.addEventListener('click', () => {
  count++;
  likeCountSpan.textContent = count;
  localStorage.setItem(storageKey, count);
});
```

- `localStorage.getItem(storageKey)` : Retrieves saved likes from the browser.
- `count++` : Increments the like counter by 1 on every click.
- `localStorage.setItem(storageKey, count)` : Saves the new number so it persists even when the user refreshes or closes the tab.

3. The Comment Submission Logic

```
commentForm.addEventListener('submit', (event) => {
  event.preventDefault(); // Stops the page from refreshing

  const name = authorInput.value.trim();
  const text = messageInput.value.trim();

  if (name !== '' && text !== '') {
    displayComment(name, text); // Adds the comment to the page
    saved.push({ name, text });
    localStorage.setItem(commentsKey, JSON.stringify(saved));
    authorInput.value = '';
    messageInput.value = '';
  }
});
```

- `event.preventDefault()` : Prevents the default browser action of reloading the page upon form submit.
- Creates a new `<div class="comment-item">` element and appends it to `#comments-list` .
- Saves the list of comments as a JSON array in `localStorage` .

4. Quick Answers for Your Presentation

Possible Lecturer Question	Your Direct Answer
"Why do the pages look so completely different?"	"Each member of Group 7 picked a distinct real-world UI design archetype. Ernest used an Instagram profile and feed, Daniella used a Notion digital workspace, Ibrahim used LinkedIn, Ntwari used an ESPN match-center card, Grace used a software developer portfolio, and Matthew used a chess/basketball court layout. We then united them using a shared navigation bar."
"Where are your icons coming from?"	"We linked the official Font Awesome CDN via a standard `<link>` tag in the `<head>` and used Font Awesome vector icon classes instead of emojis."
"How does your comment box work without a backend server?"	"We used the browser's built-in `localStorage` API. When someone posts a comment, JavaScript intercepts the form submit with `event.preventDefault()` , dynamically inserts the comment into the page DOM, and stores the list in browser memory as JSON."